

On the Invariant Theory of Forms in n Variables

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§ 2. Representation by Matrix Products.

In what follows, by a “matrix product” we always mean the sum of the products of corresponding determinants of two matrices having the same number of rows, where the rows of the second matrix are contragredient to those of the first. The rows of either matrix may be: (1) given separately; (2) combined, as in (6), into a single row S^ρ ; or (3) united into several rows $S^\rho, S^\sigma, \dots$. We denote the matrix product by the symbol $(\mid)$ used in (6). Representation (6) is a special application of a second form of the theorem on corresponding matrices: “To every ρ -row matrix product $(v^{(1)}v^{(2)} \dots v^{(\rho)} \mid y^{(1)} \dots y^{(\rho)})$ there corresponds a determinant; conversely, to every determinant there can be assigned a matrix product with any prescribed number ρ of rows ($\rho = 1, 2, \dots, n-1$).”¹ Indeed, one need only combine the y -rows into a row p_ρ (or the v -rows into a row q_ρ) and perform the substitutions given by (1), in order to pass from a given matrix product to the determinant and conversely. The value of the determinant is admittedly not given by its individual elements, but emerges only through Laplace expansion. Since determinant and matrix product thus prove equivalent, the theorem on the symbolic representability of forms (the first fundamental theorem) can also be expressed as follows.

Theorem I. “Every invariant construction can be represented symbolically by matrix products, and conversely every matrix product is an invariant construction.”²

Representation by matrix products makes it possible to overcome the essential difficulty of non-explicit representation imposed by determinant representation.³ By an “explicit representation” we mean one into which only the rows $S^\rho, T^\tau, \dots$ themselves enter, or rows R^ρ uniquely derivable from them, while the individual component rows $s, t, \dots$ no longer occur. In §6 we shall obtain, for all invariant constructions depending only on the symbol rows $S^\rho, T^\tau, \dots$, a representation explicit in those symbol rows, and thereby extend the generating operations—the contraction and differentiation operations—to the n -ary domain. The possibility of explicit representation follows because only the simplest matrix products can correspond directly to the determinants arising in determinant representation; all other matrix products therefore combine several determinants into a single expression. For if the symbol and variable rows are resolved into the individual rows s, u, x , the first fundamental theorem in its usual form necessarily yields an aggregate of determinants. The proof of the converse—that every aggregate of determinants representing an invariant construction of prescribed ground forms can be combined into explicit matrix products—uses the second fundamental theorem (§6).

To obtain from two symbol rows S^σ, T^τ , with the aid of variable rows, an invariant construction of the ground form $(S^\sigma \mid p_\sigma)(T^\tau \mid p_\tau)$ that contains all rows explicitly (the contraction operation), we need only form a matrix product of the third kind according to the definition at the beginning of this section:

$$(q_{n-\sigma-\lambda}S^\sigma \mid T_{n-\tau}p_{\tau-\lambda}), \quad (\lambda = 1, 2, \dots, \tau, n - \sigma^4). \quad (7)$$

Expanded, (7) gives

$$\begin{aligned} & (v^{(1)} \dots v^{(n-\sigma-\lambda)} s^{(1)} \dots s^{(\sigma)} \mid \alpha^{(1)} \dots \alpha^{(n-\tau)} y^{(1)} \dots y^{(\tau-\lambda)}) \\ &= \sum \left| \begin{array}{ccc} v_{i_1}^{(1)} & \dots & v_{i_{n-\lambda}}^{(1)} \\ \vdots & & \vdots \\ v_{i_1}^{(n-\sigma-\lambda)} & \dots & v_{i_{n-\lambda}}^{(n-\sigma-\lambda)} \\ s_{i_1}^{(1)} & \dots & s_{i_{n-\lambda}}^{(1)} \\ \vdots & & \vdots \\ s_{i_1}^{(\sigma)} & \dots & s_{i_{n-\lambda}}^{(\sigma)} \end{array} \right| \left| \begin{array}{ccc} \alpha_{i_1}^{(1)} & \dots & \alpha_{i_{n-\lambda}}^{(1)} \\ \vdots & & \vdots \\ \alpha_{i_1}^{(n-\tau)} & \dots & \alpha_{i_{n-\lambda}}^{(n-\tau)} \\ y_{i_1}^{(1)} & \dots & y_{i_{n-\lambda}}^{(1)} \\ \vdots & & \vdots \\ y_{i_1}^{(\tau-\lambda)} & \dots & y_{i_{n-\lambda}}^{(\tau-\lambda)} \end{array} \right| \\ &= \sum_{i,k,l} \varepsilon q_{k_1 \dots k_{n-\sigma-\lambda}} S^{k_{n-\sigma-\lambda+1} \dots k_{n-\lambda}} T_{l_1 \dots l_{n-\tau}} p^{l_{n-\tau+1} \dots l_{n-\lambda}}. \end{aligned} \quad (8)$$

¹For $\rho > n$ the matrix product is known to vanish; for $\rho = n$ it becomes the product of two determinants.

²Invariant constructions of prescribed ground forms are naturally only those aggregates of matrix products that depend on the symbol rows $S^\rho, \dots$

³Cf. the Introduction.

⁴The number λ always runs up to the smaller of the last two numbers given.

Thus it is an expression depending only on the rows S, T, q, p . Here $\varepsilon = \pm 1^5$, and the sum is to be understood so that the indices in each row $S, \dots$ are ordered among themselves, while for each combination $i_1, i_2, \dots, i_{n-\lambda}$ the k 's and the l 's separately run through all remaining possible permutations of the numbers i . When $\sigma \geq \tau$, we call the number λ entering (7) the *defect of the contraction*; the reason for this restriction will emerge in §6.

The determinants of the two matrices entering (7) may be regarded as the elements of new rows $Q^{n-\lambda}, P_{n-\lambda}$, with corresponding rows Q_λ, P^λ . By (8), these rows Q, P can be expressed in terms of the original S, q and T, p , respectively. We indicate this by

$$\begin{aligned} q_{n-\sigma-\lambda} S^\sigma &= Q^{n-\lambda} \sim Q_\lambda, & Q_\lambda &= [q_{n-\sigma-\lambda} S^\sigma]_\lambda, \\ T_{n-\tau} p_{\tau-\lambda} &= P_{n-\lambda} \sim P^\lambda, & P^\lambda &= [T_{n-\tau} p_{\tau-\lambda}]^\lambda. \end{aligned} \quad (9)$$

Taking (4) into account, the matrix product (7), analogously to (6), then has the equivalent fourfold symbolic representation

$$(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}) = (q_{n-\sigma-\lambda} S^\sigma P^\lambda) = (Q_\lambda T_{n-\tau} p_{\tau-\lambda}) = (-1)^{\lambda(n-\lambda)} (P^\lambda \mid Q_\lambda). \quad (10)$$

A further substitution of new rows can be made in (7) by setting

$$(q_{n-\sigma-\lambda} S^\sigma \mid T_{n-\tau} p_{\tau-\lambda}) = (R^{\sigma+\lambda} \mid p_{\sigma+\lambda}) (R^{\tau-\lambda} p_{\tau-\lambda}). \quad (11)$$

Here too, (8) shows that products of elements of the rows R can be expressed uniquely in terms of the rows S and T . The substitutions (9) and (11) are especially useful when (7) is to be contracted further with other rows.

The matrix representation enables us to state invariantly the quadratic relations among the elements of a row p_ρ (from which, as is known, all the others follow), hence by §1 the relations linear in the coefficients of the ground form that are satisfied by the rows S^ρ . They are the identities

$$(q_{n-\rho-\lambda} S^\rho \mid S_{n-\rho} p_{\rho-\lambda}) = 0^6, \quad (\lambda = 1, 2, \dots, \rho, n - \rho). \quad (12)$$

Here p and q are to be regarded as arbitrary quantities. In §5 we shall see that the identities with odd defect λ follow from those of higher defect. Relations (12) follow directly from relations (5), since

$$(q_{n-\rho-\lambda} S^\rho \mid S_{n-\rho} p_{\rho-\lambda}) = (v^{(1)} \dots v^{(n-\rho-\lambda)} s^{(1)} \dots s^{(\rho)} \mid \sigma^{(1)} \dots \sigma^{(n-\rho)} y^{(1)} \dots y^{(\rho-\lambda)}),$$

and application of the product theorem gives, by (5), a vanishing determinant of $n - \lambda$ rows. Identities (12) state that, to find all types of contractions, it suffices to contract the product of forms linear in each variable row; or equivalently, by §6, that the normalized form $(S^\rho \mid p_\rho)^m$ has no invariant construction linear in the coefficients.

Conditions (12) are also sufficient to characterize the rows p_ρ . Indeed, the property that the individual elements of p_ρ are determinants of a matrix is invariant and must therefore admit an invariant expression. But by Theorem VI conditions (12) impose the greatest possible invariant restriction on the p_ρ : namely, the linear form whose coefficients are the elements of the row p_ρ has no invariant constructions at all.

⁵This notation will be retained throughout.

⁶Cf. an addendum by Study in Grassmann's *Werke*, vol. I, part 2, p. 510 (the condition that a quantity S^ρ be simple).